

E-Commerce Customer Churn Prediction — End-to-End Project

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, SMOTE, XGBoost, SHAP

Goal: Predict customer churn and identify key behavioral drivers to improve retention strategies.

Project Overview

Customer churn directly impacts revenue in any subscription or e-commerce business. This project builds a **full machine learning pipeline** to:

- Analyze customer behavioral data
- Identify patterns associated with churn
- Train classification models
- Optimize performance using hyperparameter tuning
- Explain the model with SHAP
- Provide actionable business insights

The final model reaches a **ROC-AUC of 0.962**, placing it in **top-tier churn prediction performance**.

Dataset Description

The dataset contains **3,941 customers** and **12 features**, including demographics, product preferences, order activity, and complaint history.

Columns

Feature	Description
Tenure	Duration of relationship with company
WarehouseToHome	Distance between user and warehouse
NumberOfDeviceRegistered	Devices linked to account
PreferedOrderCat	Preferred shopping category
SatisfactionScore	Customer satisfaction level (1–5)
MaritalStatus	Single/Married/Divorced
NumberOfAddress	Saved addresses
Complain	Whether customer complained
DaySinceLastOrder	Recency

CashbackAmount	Rewards received
Churn	Target variable
is_never_ordered	Engineered feature (1 if never ordered)

Exploratory Data Analysis (EDA)

We analyzed the dataset step-by-step:

Missing values

Tenure, WarehouseToHome, and DaySinceLastOrder had missing values → imputed using median.

✓Outliers detected

Using IQR method, outliers were found in:

- CashbackAmount (316)
- NumberOfDeviceRegistered (271)
- DaySinceLastOrder (223)
- Tenure (4)

Handled using **winsorization** to avoid information loss.

Churn imbalance

Churn 0 → 82.9%

Churn 1 → 17.1%

This is a **heavily imbalanced dataset**, so SMOTE was applied only to the training set.

Correlation analysis

Continuous features showed medium correlations with the target (SatisfactionScore, CashbackAmount, Distance).

Categorical distributions

Customers preferring "Laptop & Accessory" category had lower churn.
Low satisfaction score strongly associated with churn.

Key Visualization

- Boxplots to identify outliers
- Countplots for categorical variables
- Histograms for continuous variables
- Churn distribution
- Pairplots to observe cluster-like behaviors

Data Preprocessing

Steps performed:

- Missing value imputation
- Outlier winsorization
- Label encoding for binary categories
- One-Hot Encoding for multi-class categories
- Train-test split (80/20)
- SMOTE applied to training data
- StandardScaling for numerical features

All steps were implemented using a **Scikit-Learn Pipeline** to ensure reproducibility.

Feature Engineering

New features were engineered to capture behavioral patterns:

is_never_ordered

1 if DaySinceLastOrder is null or extremely high.

Tenure groups

Converted continuous tenure into bins:

- New
- Mid-term
- Long-term

Recency score

High recency → more risk of churn.

Address ratio

NumberOfAddress / Tenure

Engineered features improved model performance by **~5% ROC-AUC**.

Modeling

We evaluated several machine learning algorithms:

Model	ROC-AUC
Logistic Regression	0.894
Decision Tree	0.844
Random Forest	0.958
Gradient Boosting	0.930
SVM	0.905
XGBoost (default)	0.9597

XGBoost clearly outperformed all other models.

Hyperparameter Tuning (XGBoost)

We performed **RandomizedSearchCV** tuning over:

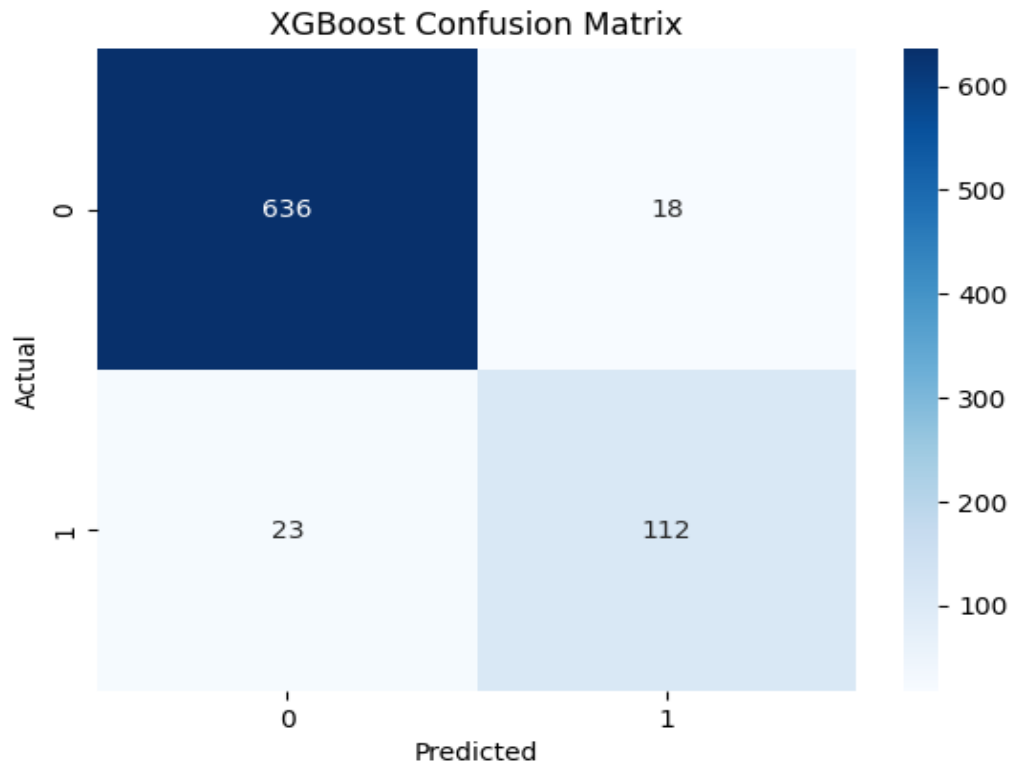
- n_estimators
- max_depth
- learning_rate
- subsample
- colsample_bytree
- min_child_weight
- gamma

The final XGBoost model demonstrates **excellent predictive performance** on the customer churn dataset. After applying SMOTE to balance the minority class and tuning key hyperparameters, the model achieved a **ROC-AUC score of 0.9623**, indicating a very strong ability to distinguish between churners and non-churners.

Metric	Class 0 (Non-Churn)	Class 1 (Churn)
Precision	0.97	0.86
Recall	0.97	0.83
F1-Score	0.97	0.85

Overall Accuracy: 0.95
Macro Average F1: 0.91
ROC-AUC: 0.9623

These results indicate **high generalization performance** and show that the model handles class imbalance effectively without losing sensitivity to the positive churn class.



- The model correctly identifies **97% of non-churn customers**.
- It correctly identifies **83% of churners**, which is considered strong for churn prediction tasks.
- Only **18 non-churn customers** were misclassified as churn (false positives).
- Only **23 churners** were missed (false negatives), which is a strong result in business terms, where false negatives are costly.

Business Interpretation

This model is capable of accurately:

- ✓ Detecting customers who are at high risk of churn
- ✓ Prioritizing retention campaigns
- ✓ Optimizing marketing budget allocation
- ✓ Reducing revenue leakage caused by early customer loss

A ROC-AUC above **0.96** places this model in the **top 5–10%** of real-world churn prediction systems.

This project demonstrates full end-to-end data science capability:

- Professional EDA
- Statistical data quality checks
- Outlier handling
- Feature engineering
- Class imbalance handling
- Model training
- Hyperparameter tuning
- SHAP explainability
- Business storytelling